

Proposed Open Agentic Interface (OAI) Specification

Standardized schemas model Accellera / IEEE 1666 SystemC reference design contracts

SCHEMA 1: FORMULATION

1. Design problem formulation

Explicit search bounds & constraints

- Swept bounds: PE array 16x16 to 64x64, SRAM capacity 256 KB to 2 MB
- System obligations: 3 W TDP limit, 120 FPS frame rate constraint
- ISA & interconnect targets: RISC-V RV64GCV, UCle die-to-die, CXL expansion
- Technology node targets: ASAP7 7 nm, TSMC N7 and N3 physical signoff bounds

SCHEMA 2: LINEAGE

2. Attempt lineage header

SLSA cryptographic provenance

- Cryptographic hash of input prompt, parent git commit, and random seed
- Multi-level IR representations: AST, CDFG, MLIR/CIRCT dialects
- Container image digest tracking: `docker://arch2/tools@sha256:a5e8f9...`
- Tool CLI arguments & UPF (IEEE 1801) power domain state declarations

SCHEMA 3: CONTRACT

3. Environmental contract

Tool AI-readiness certification

- Input prerequisites, output guarantees and multi-fidelity error bounds
- Certified simulators & EDA tools: gem5, Verilator, SCALE-Sim, Ramulator, OpenROAD, Yosys, and OpenSTA
- AI-Ready container interface features: Protobuf state, checkpoint recovery, machine-parsable error channels

Cross-organizational knowledge transfer without exposing proprietary IP

Standardized schemas allow design teams to exchange structured attempt traces safely, establishing a shared substrate for pre-trained hardware surrogates without revealing commercial foundry PDK non-disclosure parameters or proprietary RTL.